Distilling the Prompt Pathway

NeuroVestigo — closing the frame-0 conditioning gap in SAM 2 video propagation

What

SAM 2 segments the first frame of a clip with the ground-truth mask supplied as a *dense prompt*, and every later frame with no prompt at all — memory cross-attention is their only spatial context. The two regimes are not equally good. We train the no-prompt pathway to reproduce what the prompted pathway already achieves, using the model itself as the teacher.

Why — the measurement that motivates it

Per-frame validation metrics, shipped 16-frame model, best epoch (110-epoch run, 5 validation recordings):

	frame 0 (prompted)	frames 1–15 (memory only)	gap
IoU	0.9466	∼0.830	−12%
Betti-0 error	0.080	∼0.70	∼ **9**×

The topology gap is an order of magnitude larger than the region gap: losing the prompt does not blur the mask, it **fragments neurites**. And frames 1→15 drift only **0.017 in total**, so this is a *single conditioning step*, not accumulating propagation error — there are two regimes, prompted and not, and every unprompted frame sits at the same level. Improving that pathway lifts all fifteen frames together.

The decoder can produce a 0.947 mask of these images. It cannot reach that without a spatial prior.

Definitions

Let f_θ be the network, x_t frame t, m_t its ground-truth mask, $M_{<t}$ the memory bank, π the dense prompt. Teacher and student are the **same weights on the same image**, differing only in π:

$$\underbrace{\tilde{y}_t = f_\theta(x_t,\ \pi = m_t)}_{\text{teacher: privileged}} \qquad \underbrace{\hat{y}_t = f_\theta(x_t,\ \pi = \varnothing,\ M_{<t})}_{\text{student: inference-time}}$$

$$\mathcal{L}_{\text{KD}} = \lambda_1\, \mathcal{L}_{\text{Dice}}(\hat{y}_t,\ \text{sg}[\sigma(\tilde{y}_t)]) + \lambda_2\, \mathcal{L}_{\text{cbDice}}(\hat{y}_t,\ \text{sg}[\sigma(\tilde{y}_t)])$$

sg[·] is the stop-gradient. The Dice target is *soft*, $\sigma(\tilde{y}_t) \in [0,1]$: it encodes which pixels the model finds genuinely ambiguous, which a binary mask cannot. Prompt dropout withholds frame 0's prompt with probability $p_e = p \cdot \min(1, e/E_w)$, with $p = 0.4$ and $E_w = 40$.

Where the two mechanisms meet. Which frames are distilled depends on whether the prompt was dropped:

$$\mathcal{D} \;=\; \underbrace{\{1\}}_{\text{always}} \;\cup\; \underbrace{\{0 \text{ iff } \pi_0 = \varnothing\}}_{\text{only when dropped}} \;\cup\; \underbrace{\text{sample}(k,\ \{2,\dots,T-1\})}_{k=4}$$

Frame 0 is excluded whenever it keeps its prompt, because then student and teacher receive the same input and $\hat{y}_0 = \tilde{y}_0$ exactly — the term is identically zero and the decoder pass is wasted. Dropout is what puts frame 0 into the student's regime in the first place, and therefore what makes it worth a teacher at all. This is the only place the two additions interact.

The gradient, and why λ decides the outcome

Differentiating through the stop-gradient leaves **one** path — the student's:

$$\nabla_\theta \mathcal{L}_{\text{KD}} = \frac{\partial \mathcal{L}_{\text{KD}}}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial \theta}, \qquad \text{no term through } \tilde{y}_t.$$

But θ is *shared*, so the target $\text{sg}[\sigma(f_\theta(x,m))]$ is itself a function of the θ being updated. Two consequences.

(i) It is not minimising a function. With the stop-gradient the update is not ∇ of any scalar in θ: this is a dynamical system, not an optimisation problem, with no energy descended and no convergence guarantee. (Structurally the situation of BYOL / SimSiam, which need an EMA or predictor to stay stable.)

(ii) $\mathcal{L}_{\text{KD}}$ has a floor above zero. Reaching 0 would need $f_\theta(x,\varnothing,M) = f_\theta(x,m)$ for one θ — impossible, because m carries information absent from x. So $\nabla_\theta \mathcal{L}_{\text{KD}} \neq 0$ *always*, and it pushes θ forever. Writing the full objective with α for the distillation weight:

$$\mathcal{L} = \mathcal{L}_{\text{GT}} + \alpha\, \mathcal{L}_{\text{KD}} \implies \nabla \mathcal{L}_{\text{GT}}(\theta^\star) = -\alpha\, \nabla \mathcal{L}_{\text{KD}}(\theta^\star) \neq 0, \qquad \boxed{\theta^\star - \theta_{\text{GT}} \approx -\alpha\, H^{-1} \nabla \mathcal{L}_{\text{KD}}}$$

with H the Hessian of $\mathcal{L}_{\text{GT}}$. **You converge somewhere that is not the ground-truth optimum, displaced in proportion to α.** That is not a flaw to remove but a dial to set: part of $\nabla \mathcal{L}_{\text{KD}}$ points toward shapes the architecture can actually represent (useful regularisation), and part points at the unreachable residual (bias, linear in α). Small α: benefit dominates. Large α: bias dominates.

Run 1 — measured, and what it cost

Mistake	Evidence	Fix
α far too large	the two terms were **36% of** `core_loss` for the whole run	λ_1, λ_2 : 1.0/0.5 → 0.2/0.1, giving ≈10% — displacement shrinks 5×
Target still moving	teacher +0.072 IoU over epochs 2–16 vs student +0.024 over the *whole* run; gap widened 0.039 → 0.092	start at epoch 40, where the teacher is stationary (+0.002 after epoch 30). Same benefit as freezing it, no second copy of θ
Frame 1 left to chance	sampled 4 of 16 at random, so the one frame that fails got ∼25% of the budget	frame 1 **always** distilled, +4 random from frames 2+, + frame 0 when its prompt was dropped

Result: **val IoU 0.8340** against a **0.8377** baseline, then flat for 44 epochs — the signature of converging to the optimum of the wrong objective, exactly as the boxed equation predicts.

Why frame 1 and not any other. The memory bank is causal: frame 1's prediction enters $M_{<t}$ for every later t, so $\partial\mathcal{L}(t)/\partial(\text{quality at frame 1}) \neq 0$ for all $t > 1$, while frame 12 influences only three frames. Frame 1 has the largest downstream influence in the clip.

Provenance — what is borrowed, what is ours

Component	Source
Training a student on a teacher's *soft* outputs	Hinton, Vinyals & Dean, arXiv:1503.02531
Teacher holding information the student will never have	Lopez-Paz, Bottou, Schölkopf & Vapnik, ICLR 2016, arXiv:1511.03643
Withholding the prompt during training (exposure bias)	Bengio, Vinyals, Jaitly & Shazeer, NeurIPS 2015, arXiv:1506.03099
Teacher = the same weights, differing only in conditioning	**ours** — not a second network, not a deeper layer, not a prior generation
cbDice as the distillation target	**ours** — follows from the 9× topology gap; no paper suggests it
Anchoring on frame 1 via the causal memory argument	**ours**
The frame-0 conditioning gap as a measured defect	**ours** — the per-frame IoU/Betti split that motivates all of it
The loss functions themselves	the repo's existing Dice / cbDice. *Not* from the KD literature: Hinton's objective is temperature-scaled KL on softmax, which does not transfer to binary segmentation

Cost and how it will be judged

No new parameters: 5,557,250 trainable of 39,876,930 (13.9%), unchanged. The image encoder is shared, so each teacher frame costs one prompt-encoder + mask-decoder pass. Net compute is *below* run 1: $5 \times 70 = 350$ teacher passes against $4 \times 110 = 440$.

Read Betti-0 before IoU — the mechanism predicts topology moves first, since that is where the 9× gap is. And because distillation is off until epoch 40, the run carries its own control: epochs 1–39 are ordinary training, so a bend in the curve *at that boundary* is the effect. *Status: run 2 in progress.*